

Cracking the Cracker Barrel Peg Game

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Introduction

Many are familiar with the Cracker Barrel Peg Game (CBPG) as the most highway-accessible method of standardized testing in the East Coast and Midwestern United States [1]. The test administers a series of up to 14 decisions, resulting in an end state indicative of the subject's intelligence. The acknowledged end states are as follows [2]:

One peg remaining: "Genius"

Two pegs remaining: "Pretty Smart"

Three pegs remaining: "Dumb"

Four pegs remaining: "EQ-NO-RA-MOOOSE"

Much like the Rubix Cube, the CBPG is scarcely considered a legitimate demonstration of intelligence or capability in the year 2025 due to an abundance of solutions and preparatory documents published online. Instead, success on the CBPG is now attributed to the financial means and correlated free time that enable studying its solutions. Despite these well accepted shortcomings as a pure metric, the original administering institution insists on the CBPG's continued use and has even invested significant resources into its distribution and accessibility.

Assuming the CBPG is going to be around for a while, perhaps we can bring validity to its results through a more thorough analysis of its traditionally accepted and theoretically possible end states. This research will attempt to do such an analysis to yield appropriate and modern diagnostic conclusions from this antiquated test.

Rules & Convention

One of the most debated aspects of the CBPG is the initial state of the board. Most generally, the board is comprised of 15 holes arranged in a triangle and accepting 14 pegs. The location of the empty hole is sometimes said to be the first choice of the test. Others insist that the location of the empty hole is obligatory, though that position is not agreed upon. For the purposes of this research, the positions will be labeled 1-9 & a-f and the empty holes will be denoted by 0 (Figure 1):

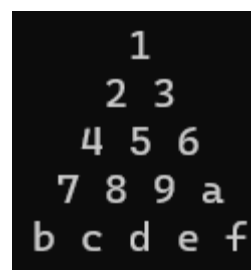


Figure 1: Peg Labels for CBPG Textual Notation

Following the initial setup, subsequent turns are taken by moving a peg across another peg, and into a hole. The "jumped" peg is removed and the board is then ready for the next turn. The objective implied by the end states is to remove as many pegs as possible.

Textual notation for the test's progress is inconsistent between solution providers, but there are two common types. The first type will be referred to as "start – destination" (SD) notation and consists of pairs of numbers representing the original location of a peg and its final location as described above. An example of SD notation is used by the test administrator's website "Move peg 4 to position 1" [2]. The alternative notation known as "source – direction" (SD) acknowledges the hexagonal grid the CBPG occupies and encodes the move as the pin to be moved and either "tl", "tr", "l", "r", "bl", "br" representing "top left", "top right", "left", "right", "bottom left", "bottom right"

respectively. An example of SD notation is “4tr”. Both examples above represent the same move:

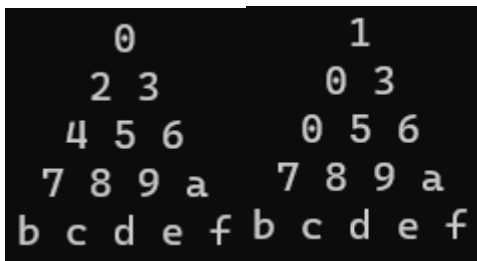


Figure 2: CBPG board state before (left) and after (right) performing the move “4tr”/“4 to 1”

In Figure 2, Peg 4 is moved to the top right and into position 1. Peg 2 is “jumped” by this action and removed.

This research will use SD notation in its descriptions of test progress. In addition, the initial hole will be denoted as simply “n” where n is the label of the starting hole.

Symmetries and States

Due to the rules of the CBPG and the arrangement of its pegs, symmetries exist that limit the number of meaningful moves. The arrangement of the board as an equilateral triangle demonstrates clearly the 3 rotational and 3 mirrored symmetries possible for a board state. Board states that can be transformed into each other through mirroring and/or rotation can be treated as equal for the purposes of analysis. Skeptics can convince themselves of this by playing in a mirror, from another chair, or some combination of these.

Returning to meaningful moves, the symmetries discussed above restrict the initial hole choice to only four categories: “1bf”, “237cae”, “46d”, and “589”. Each member of a category can be rotated and/or mirrored such that the board is equivalent to any other starting move. These categories or “symmetry groups” require only one member to be investigated to yield the next symmetry group of valid moves. Likewise, two

moves that result in the same symmetry group can be treated as the same move. For example, starting with the “1” peg removed, there are two valid moves “4tr” and “6tl”. 6tl and 4tr result in the same symmetry group of board state, meaning only one needs to be recorded. By pruning all but one member of each symmetry group at every step, we drastically reduce our search space without a loss of information.

Analysis Methods

Five meaningful variants of the CBPG exist; four obligatory starting positions (one for each symmetry group), and the variant that allows for starting choice. The starting choice variant will be investigated and the obligatory starting positions will be analyzed from their respective subset of those results.

This research will attempt to answer the question of how difficult it is to achieve the traditional end states of 1, 2, 3 and 4 pegs remaining. Due to the nature of the test, exactly one peg can and must be removed at each step, meaning games terminating in the same number of pegs must also have made the same number of moves. This would imply that all paths to an end state are truly equal in utility, lending credibility to the simple scoring used by the CBPG and eliminating the notion of an optimal path.

As an aside, it is interesting to consider how many stable states exist, and how many can be reached as a result of the test’s rules. For the purposes of this research, a statistical analysis of rarity will be used to determine “difficulty”. Presumably a result that is very uncommon will be more difficult to achieve. If a 1-peg result in a particular board state is found to be significantly rarer than a 1-peg state in another it would introduce the opportunity to further refine the diagnostic criteria of the CBPG.

Lastly, we must consider the search space required to guarantee a result. A desired outcome of 13 pegs remaining would require exactly one starting peg removal and one jump. The tester need not consider moves beyond this, severely decreasing the search space and therefore difficulty. In fact, here is exhaustive proof that a 13 peg end condition is not reachable:

1, 4tr : No
 2, 7tr : No
 2, 9tl : No
 4, btr : No
 4, dtl : No
 5, ctr : No

If a 13 peg end was possible, it certainly wouldn't have been difficult to achieve. For this reason, rarity alone is not an adequate indicator of difficulty.

In general, because the game can only progress by removing pegs, we can safely stop our search at our desired n even if the game has not terminated. In the case of $n > 1$, this $p(n)$ would introduce error because it includes -so far- identical paths that have not diverged yet. Instead, we calculate an array P indexed at n , representing the proportion of terminating states n in a search up to n . This can be obtained either by terminating the search early and flagging completion, or by processing the full data by eliminating moves after n pegs and removing duplicates.

The expected value for the number of moves per search path is represented by the following:

$$e(n) = \sum_{m=n}^{15} (15 - \max(n, m)) * P(m)$$

Eq. 1

The difficulty of a given end condition will be defined as the expected value of the number of moves made before a solution is found. If moves

are made at random, the geometric distribution's expected value gives us the following formula:

$$E(n) = \frac{1 - P[n]}{P[n]} * e(n)$$

Eq. 2

In this formula the array P indexed at n is the proportion of paths resulting in the end state with n pegs.

For our investigation, we will not be able to use this formula due to the changing probabilities caused by elimination of already searched paths. Assuming the previous moves are remembered and not taken again, the expected value is instead:

$$E(n) = \frac{O[n] + O[\sim n] + 1}{O[n] + 1} * e(n)$$

Eq. 3

Data Generation

The program performing the search used a class describing containing rules for initial moves, as well as fifteen instances of a class describing a peg/hole to represent a board state. Each peg contained references to its (up to) six neighbors, allowing for the traversal of the board as a linked graph. The class includes methods to get the valid moves, perform a move, count the remaining pegs, mirror the board, and rotate the board.

In the process of playing hundreds of thousands of games, performing deep copies of objects proved to be a significant bottleneck. To reduce object size, a new class was developed to represent the board as an array of Booleans with appropriate methods to traverse the positions. This new class introduced additional calculations for traversal that penalized many repeatedly used functions. Regardless, the change increased the speed of the search by 25% overall due to the decreased read/writes.

In the end, two additional changes in strategy resulted in a program that could complete the search. The first is the removal of one deep copy during the symmetry group check by hashing the board at the 6 rotations and mirrors, then rotating and mirroring once more to restore it to the original shape. The resulting hash also represents the entire symmetry group uniquely, giving us a rigidly defined “canonical” form to reference for every board state. The second was the use of a directed graph to associate moves and board states during the search. The resulting graph occupied less memory, ran significantly faster, and facilitated deeper analysis. Through various traversals, a complete list of unique games was generated.

Analysis of 1bf Start

In the 1bf symmetry group, it is possible to reach stable states for 1, 2, 3, 4, 5, 6, 7, and 8 pegs. Table 1 contains the number of occurrences and the proportion of games resulting in a stable end state at that many pegs. Additionally, the calculated expected value for a search with memory is included:

Table 1: Calculations for the “1bf” move

Pegs	Occurrences	Proportion	Expected
1	13005	5.24%	230.56
2	62140	25.04%	47.43
3	112797	45.44%	23.84
4	53127	21.40%	37.53
5	6614	2.66%	143.08
6	396	0.16%	773.74
7	130	0.05%	595.78
8	1	~	8123.50
Total	248210	100.00%	

A naïve analysis of the proportion of occurrences indicates that the 1-peg end state is significantly less common than 2, and 2 is less common than 3. Interestingly, the 4-peg “EQ-NO-RA-MOOOSE” end state is about as common as the 2-peg “pretty smart” state.

This data also confirms the existence of reachable end states 5, 6, 7, and 8, though they are uncommon.

The expected values for moves performed to find a solution at n depth shows a negative trend from the 1 to 3 peg states, once again supporting the end state definitions of intelligence. Like the raw proportion, the 4-peg ending is more difficult to achieve than 3, suggesting the previously accepted diagnostic criteria are not sound.

The 5 state, despite being almost twice as rare, is easier to achieve than the 1-peg. This is due to the significantly shorter search path. End states 6, 7, and 8 are more difficult than any of the explicitly acknowledged states, but still less than their miniscule proportions would imply.

Analysis of 237ace Start

The 237ace symmetry group appears similar to the 1bf symmetry group in most regards, with only minor deviations in proportion and expected values. This may be in part because moves “1 4tr” and “2 7tr” result in symmetrical boards. The only meaningfully different move “2 9tl” accounts for any difference between these symmetry groups.

Table 2: Calculations for the “237ace” move

Pegs	Occurrences	Proportion	Expected
1	13005	5.14%	234.84
2	62477	24.68%	48.07
3	114828	45.37%	23.91
4	54983	21.72%	37.15
5	7227	2.86%	135.25
6	446	0.18%	715.87
7	133	0.05%	613.72
8	1	~	8648.50
Total	253100	100.00%	

Also of interest is the 1-peg state which has an equal number of paths as the 1bf symmetry

group. An explicit analysis of the graph confirms that they are in fact the same end states. This conclusion can also be reached by an analysis of the moves following the “2 9tl” start:

Table 3: End states reachable following “2 9tl”

Pegs	Occurrences
1	0
2	337
3	2031
4	1856
5	613
6	50
7	3
8	0
Total	4890

Since this path of moves cannot result in any 1-peg end states and the 237ace symmetry group is otherwise identical to 1bf, the total 1-peg end states are the same.

Analysis of 46d Start

The 46d symmetry group is significant for its nearly doubled total of games versus the other groups discussed so far. 46d is also noteworthy for its inability to reach the 8-peg state.

Table 3: Calculations for the “46d” move

Pegs	Occurrences	Proportion	Expected
1	36957	7.15%	169.76
2	133021	25.75%	46.19
3	229247	44.38%	24.04
4	103926	20.12%	37.93
5	12550	2.43%	148.05
6	707	0.14%	870.75
7	155	0.03%	1001.69
8	0	-	-
Total	516563	100.00%	

The proportions of the 2,3,4,5, and 6 states do not differ significantly from the other symmetry groups and likewise have similar expected value

difficulties. The 5, 6, and especially 7 group are less common than the other groups, with 7 being significantly more difficult to achieve. The 1-peg state is much more common in proportion and significantly easier to achieve, though it is still more difficult than 2, 3, and 4.

So far as a standardized test this restriction of the CBPG appears to discourage the non-standard end conditions and scale the intended 1-4 peg end states more gently relative to one another.

Analysis of 589 Start

The 589 symmetry group is the smallest with less than a quarter of the paths of 1bf and 237ace, and an eighth of 46d:

Table 4: Calculations for the “589” move

Pegs	Occurrences	Proportion	Expected
1	775	1.20%	974.48
2	9838	15.28%	76.78
3	29291	45.51%	24.70
4	21621	33.59%	27.28
5	2610	4.06%	126.53
6	187	0.29%	685.20
7	41	0.06%	865.26
8	0	-	-
9	0	-	-
10	1	~	147.50
Total	64364	100.00%	

Notably, the 1-peg end state is significantly rarer in this symmetry group, also having an expected value difficulty higher than every other end state except a 7 in 46d and the 8s. The 5, 6, 7 states are also much more common than in the other symmetry groups.

This symmetry group introduces a single path to a 10-peg end state. This is the highest of any possible path, now considering all of the symmetry groups. Interestingly, although it is rare, it is relatively easy to locate due to the small

total number of paths in the 589 group and due to the shorter path required to search for it.

Analysis of Starting Choice

Looking at the case where starting pin removed is a choice, we see mostly the sum and average of the four symmetry groups we have investigated so far:

Table 5: Calculations for all starting moves

Pegs	Occurrences	Proportion	Expected
1	63742	5.89%	205.25
2	267476	24.72%	48.04
3	486163	44.92%	24.00
4	233657	21.59%	36.67
5	29001	2.68%	141.80
6	1736	0.16%	790.12
7	459	0.04%	767.10
8	2	~	24872.67
9	0	-	-
10	1	~	1185.00
Total	1082237	100.00%	

In terms of unique symmetry groups of states that can be reached at any point during the game, there are only 2545. Also, because a stable state with number 12 can be constructed by removing pegs 5, 8, and 9 but there are no games ending with 12 pegs, we can observe that the set of stable states reachable by the CBPG is not the set of all possible stable states.

Conclusions

Unsurprisingly, the 1, 2, and 3 peg states rank in difficulty in alignment with the CBPG's labels. The 4-peg state is easier to locate than the 3, suggesting that a 3 peg end state should signify an "EQ-NO-RA-MOOOSE" instead.

There is still the problem of the 5, 6, 7, 8, and 10 states. If they are to be considered, the order of difficulty is much more disrupted vs the traditional CBPG criteria. The following illustrates my suggested ordering and labels:

Table 6: Proposed New CBPG Diagnostic Criteria

Pegs	Expected	Label
8	24872.67	Cracker (of the Barrel)
9	13775.00	Liar
10	1185.00	Brilliant
6	790.12	Genius
7	767.10	Genius Jr.
1	205.25	Smart
5	141.80	Pretty Smart
2	48.04	Pretty Dumb
4	36.67	Dumb
3	24.00	EQ-NO-RA-MOOOSE

Also, although a 9-peg stable state is impossible with the rules of the CBPG, an ignorant participant with memory could rule it out with certainty in fewer moves than would be expected to find a single solution for the 8-peg end state.

The 8-peg end state remains the most difficult state by ranking and is now significantly more laborious to locate with the inclusion of the other symmetry groups. As a challenge to the reader, try to solve it without consulting this research's supporting documents. I suspect the blind and random approach used to quantify difficulty in this research is not fully representative of the strategies and optimizations an actual human subject could employ. I have refrained from viewing the solution beyond the above analysis of starting moves with the hopes that this challenge will remain interesting at my next CBPG evaluation.

Fellow academics completing a solution to the 8-peg end state without aid should submit an attestation to my nearest testing location; 4140 Pier North Blvd. Flint, MI 48504. Please address your letter to "Honorary Cracker of the Barrel, Henry Grasman" and include a return address that cannot be accessed by wild animals.

Future Work

Attempts to create a readable infographic for this game's play have ultimately failed. The current best candidate for a representation of the CBPG's states is the following 3D arrangement of nodes:

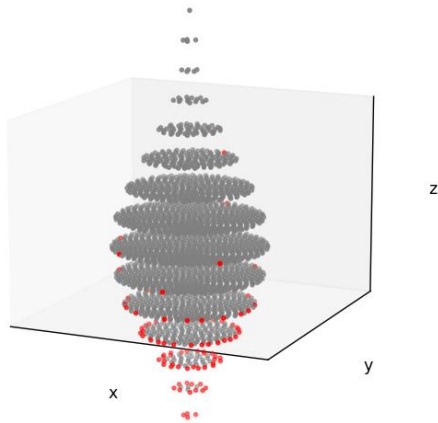


Figure 3: CBPG Network Arranged in Circular Layers without Edges

This representation has not been optimized to allow for readability once edges are drawn. The availability of volumetric displays is also scarce at this point in human history. An improvement to both could yield a usable reference for CBPG progression.

A 2D version of this has been constructed and unsuccessfully optimized to reduce edge crossings. Perhaps as a billboard-sized printout it can yield usable insight:



Figure 4: CBPG Reference in 2D

To achieve a target number end-state it is trivial with these references and the data they represent. Starting at a stable end state, the network can be traversed backward arbitrarily until the board is filled. Memorizing that single path is sufficient to achieve the desired score on the CBPG. A turn-based adversarial version of the

game provides a new and interesting challenge. The rules are as follows:

- 1) Pegs are moved according to the original CBPG rules.
- 2) Players alternate moves. The player making the first move is attempting to minimize the peg count. The second player is attempting to maximize the peg count.
- 3) After a stable state is reached, the order of the players is swapped, and the process is repeated with a new board.
- 4) The player achieving the least-peg end state of the two rounds is the winner.

The ruining of this game is not immediately obvious, and a bare minimum analysis will be required to determine if it is possible to force a tie like some antiquated roadside adversarial intelligence tests.

References

- [1] Finlayson, Caitlin & Gregory, Meredith. (2019). The Paradox of Cracker Barrel: A Case Study on Place and Placelessness. *Advances in Hospitality and Tourism Research (AHTR)*. 7. 258-276. 10.30519/ahtr.537204.
- [2] How to solve the cracker barrel Peg game: Cracker Barrel blog. Cracker Barrel Insider Blog. (2023, August 17). <https://blog.crackerbarrel.com/2021/08/13/how-to-beat-the-cracker-barrel-peg-game/>